Abstract

This is a sample research abstract discussing the application of Neural Networks to General Organic Chemistry (GOC). We explore how deep learning models can classify molecules and predict reaction pathways.

Introduction

In this paper, we investigate machine learning approaches for General Organic Chemistry (GOC). Organic chemistry relies on structural rules that can be learned by graph neural networks.

Methodology

Our proposed method represents molecules as molecular graphs. We train a GCN model to predict stability metrics.

Conclusion

We demonstrated that deep neural networks can learn fundamental GOC concepts. Future work will expand to larger reaction datasets.

References

[1] Smith, J. et al. Deep learning in organic synthesis. Journal of Chemistry, 2024.
[2] Johnson, R. Graph neural networks for molecules. arXiv, 2025.